

The P·E·D·W Worksheet

Forward + backprop by hand, one row of golf data - the ANSWER KEY

THE SETUP - ONE ROW OF THE GOLF LOGBOOK

Day D1 said: **hot, humid, calm ...and the players stayed home.**

Recorded for the network (1 = hot / humid / windy / played, 0 = cool / dry / calm / stayed home):

	Temp	Humidity	Wind		Play?
inputs $x =$	1	1	0	target $y =$	0
	w1	w2	w3		
weights $w =$	0.52	0.18	-0.12	$\alpha = 0.1$	(learning rate)

The starting weights are random guesses. Your job: use P·E·D·W three times and watch them learn.

THE FOUR MOVES - SAY THEM IN ORDER, EVERY TIME

P	PREDICTION $P = x \cdot w = (x1 \times w1) + (x2 \times w2) + (x3 \times w3)$	the dot product - multiply and add
E	ERROR $E = (y - P)^2$	squared so it is always positive (just for tracking)
D	DELTA $D = P - y$	signed miss: + means we OVERestimated
W	WEIGHT UPDATE new weight = old weight - $\alpha \cdot (\text{its input} \times D)$	scale by the input, step against the miss

Overestimated ($D > 0$)? weights SHRINK. Underestimated ($D < 0$)? weights GROW. Input = 0? that weight does NOT move!

HOW TO USE THIS KEY

Every rectangle and every formula is filled in. Follow one round top to bottom, then try the blank version yourself - same numbers, no peeking until you finish a round.

ROUND 1

	Temp	Hum	Wind
$x =$	1	1	0

	w1	w2	w3
$w \text{ in} =$	0.52	0.18	-0.12

P $(1 \times 0.52) + (1 \times 0.18) + (0 \times -0.12) = 0.70$

E $(0 - 0.70)^2 = 0.49$

D $0.70 - 0 = +0.70$

W $w1 = 0.52 - 0.1 \times (1 \times 0.70)$ $w2 = 0.18 - 0.1 \times (1 \times 0.70)$ $w3 = -0.12 - 0.1 \times (0 \times 0.70)$

$w \text{ out} =$	0.45	0.11	-0.12
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← w3 didn't move - its input was 0!

ROUND 2

	Temp	Hum	Wind
$x =$	1	1	0

	w1	w2	w3
$w \text{ in} =$	0.45	0.11	-0.12

P $(1 \times 0.45) + (1 \times 0.11) + (0 \times -0.12) = 0.56$

E $(0 - 0.56)^2 = 0.3136$

D $0.56 - 0 = +0.56$

W $w1 = 0.45 - 0.1 \times (1 \times 0.56)$ $w2 = 0.11 - 0.1 \times (1 \times 0.56)$ $w3 = -0.12 - 0.1 \times (0 \times 0.56)$

$w \text{ out} =$	0.394	0.054	-0.12
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← carry these into Round 3

ROUND 3

	Temp	Hum	Wind
$x =$	1	1	0

	w1	w2	w3
$w \text{ in} =$	0.394	0.054	-0.12

P $(1 \times 0.394) + (1 \times 0.054) + (0 \times -0.12) = 0.448$

E $(0 - 0.448)^2 = 0.2007$

D $0.448 - 0 = +0.448$

W $w1 = 0.394 - 0.1 \times (1 \times 0.448)$ $w2 = 0.054 - 0.1 \times (1 \times 0.448)$ $w3 = -0.12 - 0.1 \times (0 \times 0.448)$

$w \text{ out} =$	0.3492	0.0092	-0.12
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← final weights!

WHAT TO NOTICE: P marches toward the target: $0.70 \rightarrow 0.56 \rightarrow 0.448$ (each round is $0.8 \times$ the last!)

We overestimated every round ($D > 0$), so the active weights kept shrinking - and w3 never moved. Zero in, zero learned.